

Applications of Differentiation

(1) Calculation of Approximation (Differentials)

A method for approximating the value of a function **near a known value**. In this method Δx and Δy represent the **changes** in x and y for the function, and dx and dy represent the **changes** in x and y **for the tangent line**.

$$dy = f'(x)dx$$

$$\Delta x = dx$$

$$\Delta y = dy = f'(x)\Delta x$$

Therefore, we have

$$\begin{aligned} f(x + \Delta x) &= f(x) + \Delta y \\ &= f(x) + f'(x)\Delta x \end{aligned}$$

Example: Using differentials, calculate each of the following values

(1) Approximate $\sqrt{10}$

Note that $\sqrt{10}$ is near $\sqrt{9}$; so we will use

$$f(x) = \sqrt{x} \text{ with } x = 9 \text{ and } \Delta x = 1; f'(x) = \frac{1}{2\sqrt{x}}$$

$$\begin{aligned} \sqrt{10} &= f(x + \Delta x) \\ &\approx f(x) + f'(x)\Delta x \\ &= \sqrt{x} + \frac{1}{2\sqrt{x}}(1) = \sqrt{9} + \frac{1}{2\sqrt{9}} = 3 + \frac{1}{6} = 3.1666 \end{aligned}$$

(2) Approximate $(995)^{\frac{1}{3}}$

Note that $(995)^{\frac{1}{3}}$ is near $(1000)^{\frac{1}{3}}$; so we will use

$$f(x) = \sqrt[3]{x} \text{ with } x = 1000 \text{ and } \Delta x = -5; f'(x) = \frac{1}{3} x^{-\frac{2}{3}}$$

$$\begin{aligned} (995)^{\frac{1}{3}} &= f(x + \Delta x) \\ &\approx f(x) + f'(x)\Delta x \\ &= \sqrt[3]{x} + \frac{1}{3} x^{-\frac{2}{3}}(-5) = \sqrt[3]{1000} + \frac{-5}{300} = 10 - 0.0166 = 9.9833 \end{aligned}$$

(3) Approximate $\tan(46)$

Note that $\tan(46)$ is near $\tan(45)$, so we will use

$$f(x) = \tan(x) \text{ with } x = 45^\circ = \frac{\pi}{4} \text{ and } \Delta x = 1^\circ = \frac{\pi}{180}; f'(x) = \sec^2(x) = 2$$

$$\begin{aligned} \tan(46) &= f(x + \Delta x) \\ &\approx f(x) + f'(x)\Delta x \\ &= \tan\left(\frac{\pi}{4}\right) + \sec^2\left(\frac{\pi}{4}\right)\left(\frac{\pi}{180}\right) = 1 + 2 * \frac{\pi}{180} = 1.0349 \end{aligned}$$

Exercise (8-1)

Using differentials, calculate each of the following values

(1) $y = \sqrt{5}$

(2) $y = \sin(29)$

(3) $y = \tan(43)$

(4) $y = \sqrt{50}$

(5) $y = \sqrt{26}$

(6) $y = \frac{1}{99}$

(2) Tangents and Normals

A **tangent** to a curve is a line that touches the curve at one point and has the same **slope** as the curve at that point. A **normal** to a curve is a line **perpendicular** to a tangent to the curve.

* The **equation of the tangent** is given by

$$y - y_1 = m(x - x_1)$$

* The **equation of the normal** is given by

$$y - y_1 = -\frac{1}{m}(x - x_1)$$

where $m \equiv \frac{dy}{dx}$ is the **slope** of the tangent at the point (x_1, y_1)

Example

Find the equation of the tangent and normal to the curve $4x = y^2$ at the point $(2,1)$.

Solution

First, we find the **slope** of the tangent at the point (x_1, y_1)

$$y^2 = 4x \quad \Rightarrow \quad 2y \frac{dy}{dx} = 4 \quad \Rightarrow \quad \frac{dy}{dx} = \frac{4}{2y} = \frac{4}{2 \times 1} = 2$$

The **equation of the tangent** is

$$y - y_1 = (2)(x - x_1)$$

$$y - 1 = (2)(x - 2) \Rightarrow 2x - y - 3 = 0$$

The **equation of the normal** is

$$y - y_1 = -\frac{1}{m}(x - x_1)$$

$$y - 1 = \left(-\frac{1}{2}\right)(x - 2) \Rightarrow x + 2y - 4 = 0$$

Exercise (8-2)

Find the equation of the tangent and normal to the following curves with corresponding points

(1) $x^2 = 4y$; $(2,1)$

(2) $y = x^2 + x + 1$; $(1,3)$

(3) $x^2 = 3y + 3$; $(3,2)$

(4) $x^2 = 4y - 3$; $(5,7)$

(3) L'hospital's Rule

Suppose that we have one of the following **indeterminate cases**,

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{0}{0} \quad \text{or} \quad \lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{\infty}{\infty}$$

Where a can be any real number, in these cases we have

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$$

L'hospital's Rule can help us to calculate a limit that may otherwise be **hard** or **impossible**.

Example: Using L'hospital's rule to find limits of the following

1. $\lim_{x \rightarrow 2} \frac{x^3 - 8}{x^2 - 4} = \lim_{x \rightarrow 2} \frac{3x^2}{2x} = \frac{4 \times 4}{2 \times 2} = 3$
2. $\lim_{x \rightarrow 3} \frac{2^x - 2^3}{x - 3} = \lim_{x \rightarrow 2} \frac{2^x \ln(2)}{1} = 8 \ln(2)$
3. $\lim_{x \rightarrow 0} x \ln(x) = \lim_{x \rightarrow 0} \left(\frac{\ln(x)}{\frac{1}{x}} \right) = \lim_{x \rightarrow 0} \left(\frac{\frac{1}{x}}{-\frac{1}{x^2}} \right) = \lim_{x \rightarrow 0} -x = 0$
4. $\lim_{x \rightarrow 0} \left(\frac{e^{2x} - 1}{x} \right) = \lim_{x \rightarrow 0} \left(\frac{2e^{2x}}{1} \right) = 2$
5. $\lim_{x \rightarrow \infty} \frac{x^2}{e^x} = \lim_{x \rightarrow \infty} \frac{2x}{e^x} = \lim_{x \rightarrow \infty} \frac{2}{e^x} = \frac{2}{\infty} = 0$

Exercise (8-3)

Find the values for each of the following limits

- | | | |
|---|---|---|
| (1) $\lim_{x \rightarrow \infty} \left(\frac{3x^2 - x + 5}{5x^2 + 6x - 3} \right)$ | (2) $\lim_{x \rightarrow 1} \left(\frac{-\pi \sin(\pi x)}{2x - 2} \right)$ | (3) $\lim_{x \rightarrow 0} \left(\frac{\ln(\cos(3x))}{\ln(\cos(2x))} \right)$ |
| (4) $\lim_{x \rightarrow 0} \left(\frac{1}{\sin^2(x)} - \frac{1}{x^2} \right)$ | (5) $\lim_{x \rightarrow -1} \frac{\sin(x^2 - x - 2)}{x + 1}$ | (6) $\lim_{x \rightarrow 1} \frac{\sin(1 - \sqrt{x})}{x - 1}$ |
| (7) $\lim_{x \rightarrow 0} \frac{\sin(x^2)}{x}$ | (8) $\lim_{x \rightarrow 0} \frac{\sin(1 - \cos(x))}{x}$ | (9) $\lim_{x \rightarrow 2} \frac{\sin(x^2 - 4)}{x - 2}$ |
| (10) $\lim_{x \rightarrow 9} \frac{\sin(\sqrt{x} - 3)}{x - 9}$ | (11) $\lim_{x \rightarrow \infty} \left(\frac{\ln(x)}{x} \right)$ | (12) $\lim_{x \rightarrow 0} \frac{\sin(\sin(x))}{x}$ |

(4) Higher order derivatives (nth derivative)

The **nth derivative** is denoted by the symbol $\frac{d^ny}{dx^n}$ or $y^{(n)}$

Example: Find the nth derivative for each of the following functions

(1) $y = \sin(x)$

$$y' = \cos(x) = \sin\left(x + \frac{\pi}{2}\right)$$

$$y'' = -\sin(x) = \sin\left(x + \frac{2\pi}{2}\right)$$

$$y''' = -\cos(x) = \sin\left(x + \frac{3\pi}{2}\right)$$

$$y^{(4)} = \sin(x) = \sin\left(x + \frac{4\pi}{2}\right)$$

$$\therefore y^{(n)} = \sin\left(x + \frac{n\pi}{2}\right)$$

(2) $y = e^{ax}$

$$y' = ae^{ax}$$

$$y'' = a^2e^{ax}$$

$$y''' = a^3e^{ax}$$

$$\therefore y^{(n)} = a^ne^{ax}$$

(3) $y = \frac{1}{1+x}$

$$y' = (-1)(1+x)^{-2}$$

$$y'' = (-1)(-2)(1+x)^{-3}$$

$$y''' = (-1)(-2)(-3)(1+x)^{-4}$$

$$\therefore y^{(n)} = (-1)^nn!(1+x)^{-(n+1)}$$

Exercise (8-4)

1- Find the nth derivative for each of the following functions

(1) $y = \ln(1-x)$ (2) $y = xe^x$ (3) $y = x^2\ln(x)$

(4) $y = \frac{1}{1-x}$ (5) $y = \sqrt{x}$ (6) $y = (ax+b)^n$

2- If $y = e^{-2x} \sin(4x+5)$ show that $y'' + 4y' + 20y = 0$

3- If $y = \frac{1}{2}x^2e^x$ show that $y'' - 2y' + y = e^x$

4- If $y^n x^m = (x+y)^{n+m}$ show that $\frac{dy}{dx} = \frac{y}{x}$